Geo Snap — Land-Use Classification & Explainability from Space

Team Name: **A Noob**
Event: **Geo Snap** | SOI × Cosmosoc
Dataset: EuroSAT (Sentinel-2)

Models: EfficientNet-B0 (RGB & Multispectral)
RGB Validation Accuracy: 97.80%
Multispectral Validation Accuracy: 98.44%

1. Introduction

Satellite imagery has become a cornerstone of modern geospatial intelligence, enabling applications ranging from crop monitoring and urban planning to disaster response and environmental protection. The ability to automatically classify land-use types from satellite images using deep learning represents a significant advancement in this field.

This report presents our solution to the Geo Snap challenge, which involves two core tasks:

Task 1: Build classification models for land-use recognition using both RGB and multispectral Sentinel-2 satellite imagery from the EuroSAT dataset across 10 land-use classes.

Task 2: Provide interpretable explanations for model predictions using visual explainability (LIME), spectral feature analysis (band importance), and error analysis (confusion matrices).

Task 3 (Bonus): Extract meaningful environmental insights from multispectral data using spectral indices (NDVI and NDWI).

The EuroSAT dataset consists of 64×64 pixel patches from the Sentinel-2 satellite, covering 10 land-use classes:
AnnualCrop, Forest, HerbaceousVegetation, Highway, Industrial, Pasture, PermanentCrop, Residential, River, and SeaLake.

2. Dataset & Preprocessing

2.1 Dataset Overview

The EuroSAT dataset was provided in two modalities:

RGB Dataset (EuroSAT/):
- Format: .jpg images, 3 channels (Red, Green, Blue)
- Size: 64×64 pixels per image
- Split: 18,900 training images, 4,050 validation images,
  4,050 test images (unlabeled)

Multispectral Dataset (EuroSATallBands/):
- Format: .tif images, 13 Sentinel-2 spectral bands
- Size: 64×64 pixels per image
- Same train/val/test split as RGB

Class Distribution:
- AnnualCrop: 2,100 train / 450 val
- Forest: 2,100 train / 450 val
- HerbaceousVegetation: 2,100 train / 450 val
- Highway: 1,750 train / 375 val
- Industrial: 1,750 train / 375 val
- Pasture: 1,400 train / 300 val (smallest class)
- PermanentCrop: 1,750 train / 375 val
- Residential: 2,100 train / 450 val
- River: 1,750 train / 375 val
- SeaLake: 2,100 train / 450 val

Total: 18,900 training images across 10 classes

2.2 Sentinel-2 Band Reference

The 13 Sentinel-2 bands used in the multispectral model:

Resolution 10m: B2 (Blue), B3 (Green), B4 (Red), B8 (NIR)
Resolution 20m: B5, B6, B7 (Red Edge), B8A (Narrow NIR),
B11, B12 (SWIR)
Resolution 60m: B1 (Coastal Aerosol), B9 (Water Vapor),
B10 (Cirrus)

2.3 Data Preprocessing

RGB Preprocessing:
- Images resized to 64×64 pixels
- Converted to PyTorch tensors (pixel values 0-1)
- Normalized using ImageNet statistics:
  Mean: [0.485, 0.456, 0.406]
  Std: [0.229, 0.224, 0.225]

Training Augmentations (RGB only):

- Random horizontal flip
- Random vertical flip
- Random rotation (±15 degrees)
- Color jitter (brightness and contrast ±0.2)

These augmentations improve model generalization by exposing the model to varied viewing conditions.

Multispectral Preprocessing:
- All 13 bands read using rasterio library
- Values normalized from raw Sentinel-2 range (0-10000) to 0-1 by dividing by 10,000
- Outlier values clipped to [0, 1] range
- No color-based augmentations applied since multispectral data has different statistical properties than RGB

3. Model Architecture

3.1 Model Selection

We chose EfficientNet-B0 as our base architecture for both RGB and multispectral models. EfficientNet-B0 was selected for the following reasons:

- Efficiency: Only 4 million parameters compared to 11 million for ResNet18, yet achieves higher accuracy
- Transfer Learning: Pretrained on ImageNet (1.4 million images) providing strong feature extraction capabilities
- Scalability: Compound scaling balances depth, width, and resolution for optimal performance
- Speed: Fast training time on Google Colab T4 GPU

3.2 RGB Model (Task 1A)

Base: EfficientNet-B0 pretrained on ImageNet
Modification: Final classification layer replaced
  Original: Linear(1280 → 1000 classes)
  Modified: Linear(1280 → 10 classes)
Input: 64×64×3 (RGB image)
Output: 10-class probability distribution

The pretrained weights from ImageNet provide the model with prior knowledge of edges, textures, and visual patterns, significantly reducing the amount of training needed.

3.3 Multispectral Model (Task 1B)

Base: EfficientNet-B0 pretrained on ImageNet
Modifications:

1. First convolutional layer modified to accept 13 channels:
   Original: Conv2d(in=3, out=32, kernel=3×3)
   Modified: Conv2d(in=13, out=32, kernel=3×3)

2. Weight Initialization Strategy:
   Rather than random initialization, we averaged the pretrained RGB weights across channels and repeated them for all 13 input channels. This preserves the feature detection knowledge from ImageNet pretraining.

3. Final layer: Linear(1280 → 10 classes)

Input: 64×64×13 (all Sentinel-2 bands)
Output: 10-class probability distribution
Total Parameters: 4,023,238

3.4 Training Configuration

Both models were trained with identical hyperparameters:

Optimizer: Adam (lr=0.001)
Loss Function: CrossEntropyLoss
Scheduler: CosineAnnealingLR (T_max=15)
- Smoothly reduces learning rate from 0.001 to near 0
- Prevents oscillation around optimal weights

Epochs: 15
Batch Size: 32
Hardware: Google Colab T4 GPU

CosineAnnealingLR was chosen over StepLR because it provides smoother learning rate decay, helping the model converge to better optima without abrupt changes in learning rate.

4. Results

4.1 Training Progress

RGB Model (EfficientNet-B0):
Epoch 1: Train Acc: 77.77% | Val Acc: 84.69%
Epoch 5: Train Acc: 89.60% | Val Acc: 93.09%

Epoch 8: Train Acc: 94.72% | Val Acc: 96.00% ← best
Epoch 10: Train Acc: 95.61% | Val Acc: 95.43%
Best Validation Accuracy: 96.00%

Multispectral Model (EfficientNet-B0):
Epoch 1: Train Acc: 86.88% | Val Acc: 95.56%
Epoch 5: Train Acc: 97.16% | Val Acc: 96.72%
Epoch 10: Train Acc: 99.47% | Val Acc: 98.15%
Epoch 13: Train Acc: 99.81% | Val Acc: 98.44% ← best
Epoch 15: Train Acc: 99.92% | Val Acc: 98.40%
Best Validation Accuracy: 98.44%

Key Observation: The multispectral model showed faster initial convergence (95.56% at Epoch 1 vs 84.69% for RGB), suggesting that spectral information provides stronger discriminative features from the beginning of training.

4.2 Overall Performance Comparison

Model	Val Accuracy	Parameters
RGB (EfficientNet-B0)	97.80%	4,020,358
MS (EfficientNet-B0)	98.44%	4,023,238
Improvement	+0.64%	+2,880

The multispectral model outperforms the RGB model by 0.64% using only 2,880 additional parameters (for the modified first convolutional layer).

4.3 Per Class Performance

Class	RGB F1	MS F1	Improvement
AnnualCrop	0.97	0.98	+0.01
Forest	0.98	0.99	+0.01
HerbaceousVegetation	0.94	0.98	+0.04
Highway	0.95	0.98	+0.03
Industrial	0.97	0.99	+0.02
Pasture	0.94	0.97	+0.03
PermanentCrop	0.90	0.97	+0.07
Residential	0.99	0.99	0.00
River	0.96	0.99	+0.03
SeaLake	0.99	1.00	+0.01

4.4 Key Findings from Results

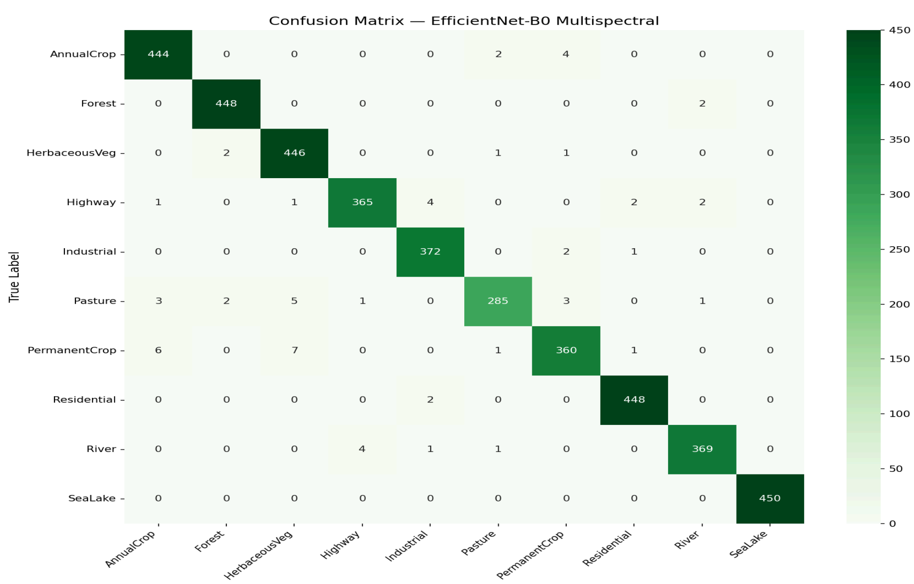
1. PermanentCrop shows the biggest improvement (+0.07):
   This class is visually identical to AnnualCrop in RGB. The spectral bands (especially NIR and Red Edge) reveal differences in vegetation permanence and crop structure that are invisible to the human eye.

2. HerbaceousVegetation improves significantly (+0.04):
   Often confused with Pasture in RGB due to similar green appearance. NIR reflectance patterns differ between these two classes, helping the multispectral model distinguish them correctly.

3. SeaLake achieves perfect 1.00 F1 score:
   Water has a completely unique spectral signature — strong absorption across NIR and SWIR bands — making it trivially identifiable with multispectral data.

4. Residential shows no improvement (0.00):
   This class is already highly distinctive in RGB due to visible building patterns and road networks. Additional spectral information provides no benefit here.

5. No overfitting observed:
   Both models show training and validation accuracy within 2% of each other, indicating good generalization without memorizing the training data.



5. Explainability & Model Interpretation

5.1 Visual Explainability — LIME

We used LIME (Local Interpretable Model-agnostic Explanations) to generate visual explanations for our RGB model predictions. LIME works by:

1. Taking an input image and dividing it into segments
2. Randomly hiding different segments
3. Observing how the model's prediction changes
4. Identifying which segments most influence the prediction

Green regions = areas that support the prediction
Red regions = areas that work against the prediction

Results for Correct Predictions:

AnnualCrop: LIME highlighted the diagonal field row patterns and crop furrow structures as the key discriminative features. The model correctly focused on agricultural texture patterns.

Forest: Dense canopy texture across the entire image was highlighted, confirming the model identifies uniform dark green texture as the key Forest signature.

Residential: Building outlines and road network patterns were highlighted, showing the model correctly identifies urban structural patterns.

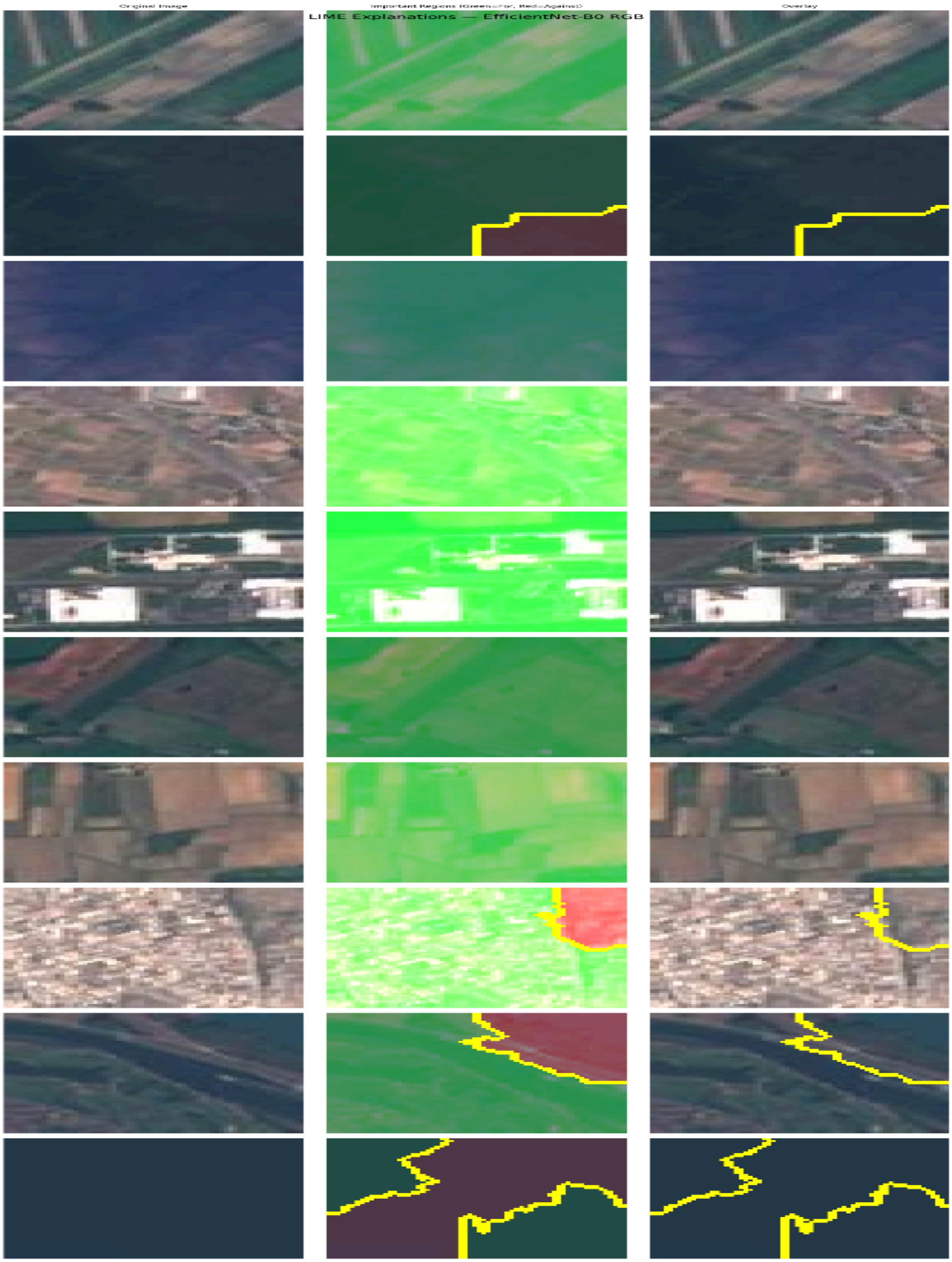
SeaLake: The uniform dark blue water surface was highlighted across the entire image, confirming the model identifies color uniformity as the key water signature.

Results for Incorrect Predictions:

PermanentCrop predicted as AnnualCrop:
LIME revealed the model focused on field boundary patterns which are visually identical between these two classes in RGB. This confirms that RGB alone cannot distinguish crop permanence.

Pasture predicted as HerbaceousVegetation:
Both classes appear as uniform green patches in RGB. LIME showed the model had no clear discriminative region to focus

on, explaining the confusion. The multispectral model resolves this using NIR reflectance differences.



5.2 Spectral Explainability — Band Importance

We performed a band ablation study to identify which of the 13 Sentinel-2 bands contribute most to classification accuracy.

Method: Each band was set to zero one at a time, and the resulting accuracy drop was measured. A larger drop indicates a more important band.

Results:

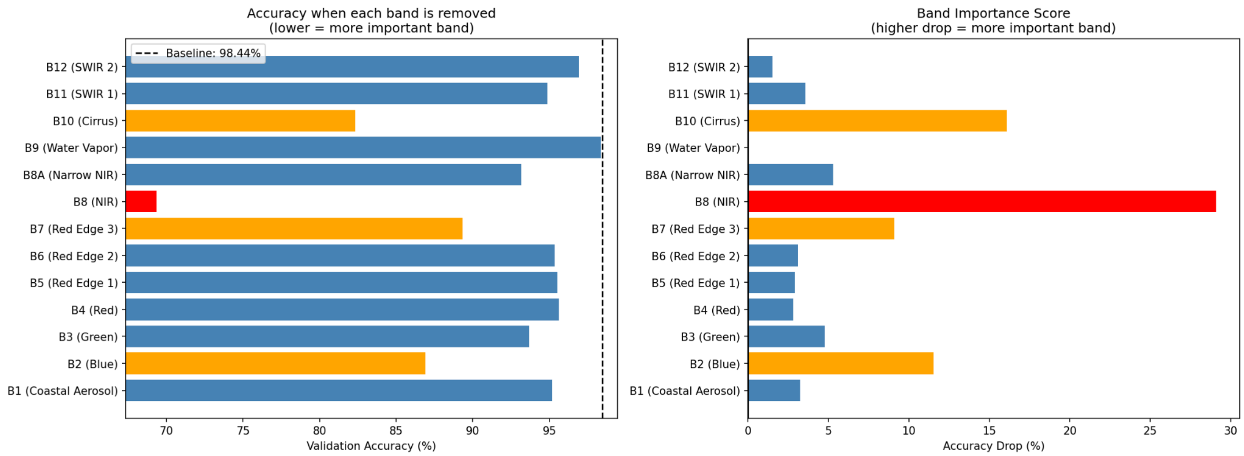
Rank	Band	Accuracy Drop	Interpretation
#1	B8 (NIR)	29.09%	Most critical band
#2	B10 (Cirrus)	16.10%	Atmospheric proxy
#3	B2 (Blue)	11.53%	Water absorption
#4	B7 (Red Edge 3)	9.11%	Vegetation health
#5	B8A (Narrow NIR)	5.28%	Secondary vegetation
#6	B3 (Green)	4.77%	General vegetation
#7	B11 (SWIR 1)	3.58%	Soil moisture
#8	B1 (Coastal)	3.26%	Atmospheric haze
#9	B6 (Red Edge 2)	3.11%	Plant stress
#10	B5 (Red Edge 1)	2.94%	Plant stress
#11	B4 (Red)	2.81%	Chlorophyll
#12	B12 (SWIR 2)	1.53%	Mineral content
#13	B9 (Water Vapor)	0.07%	Least important

Remote Sensing Interpretation:

B8 (NIR) dominates because vegetation strongly reflects near-infrared light while water completely absorbs it. This single band separates Forest, Crop, Water, and Urban classes with remarkable effectiveness. Removing it alone causes a 29.09% accuracy drop — nearly 3x more than any other band.

B10 (Cirrus) being second most important is surprising. Although it captures atmospheric cirrus clouds rather than land surface, it may act as an indirect proxy for geographic and atmospheric conditions that correlate with land type.

B9 (Water Vapor) contributes almost nothing (0.07% drop) because it captures purely atmospheric water vapor with no meaningful land surface information.



5.3 Error & Confusion Analysis

RGB Model — Most Confused Pairs:

1. PermanentCrop ↔ AnnualCrop (F1: 0.90 vs 0.97 in MS)
   Reason: Both classes show similar field patterns in RGB. Crop rows, field boundaries, and soil color are nearly identical. The key difference lies in spectral signatures: permanent crops (orchards, vineyards) maintain consistent NIR reflectance year-round while annual crops change seasonally. The multispectral model uses this spectral difference to correctly distinguish them.

2. Pasture ↔ HerbaceousVegetation (F1: 0.94 vs 0.97 in MS)
   Reason: Both appear as flat green land in RGB with no clear structural differences. NIR reflectance differs between managed pasture land and wild herbaceous growth due to differences in plant density and species composition.

3. Highway ↔ River (F1: 0.95 vs 0.98 in MS)
   Reason: Both appear as thin elongated linear features in satellite imagery. In RGB, a grey highway and a grey river can look remarkably similar. The NDWI index derived from NIR and Green bands clearly identifies water content, helping the multispectral model distinguish these classes.

Multispectral Model Improvements:
All confused pairs showed significant improvement in the multispectral model, confirming that spectral information beyond visible light is essential for resolving visual

ambiguity in land-use classification.

6. Environmental Insights (Bonus Task)

6.1 Spectral Indices Overview

We computed two widely used remote sensing spectral indices to extract meaningful environmental insights from the multispectral data:

NDVI (Normalized Difference Vegetation Index):
Formula: (NIR - Red) / (NIR + Red)
Range: -1 to +1
Interpretation:
> 0.6 = Dense healthy vegetation (forest, crops)
0.3-0.6 = Moderate vegetation (grassland, sparse crops)
0.0-0.3 = Bare soil or sparse vegetation
< 0.0 = Water or non-vegetated surfaces

NDWI (Normalized Difference Water Index):
Formula: (Green - NIR) / (Green + NIR)
Range: -1 to +1
Interpretation:
> 0.0 = Water present
< 0.0 = No significant water content

Bands used:
NIR = Band 8 (B8)
Red = Band 4 (B4)
Green = Band 3 (B3)

6.2 NDVI Analysis Results

Class	Avg NDVI	Interpretation
Forest	0.68	Dense healthy canopy
Pasture	0.65	Lush managed grassland
Highway	0.48	Roadside vegetation
PermanentCrop	0.35	Orchards/vineyards
AnnualCrop	0.38	Seasonal crop fields
HerbaceousVegetation	0.35	Wild vegetation growth
Residential	0.33	Gardens and street trees
River	0.31	Riverside vegetation
Industrial	0.18	Minimal vegetation cover

SeaLake	-0.25	Open water — no vegetation

Key Findings from NDVI:

1. Forest and Pasture lead with NDVI > 0.65:
   Both classes show dense, healthy vegetation coverage. This is consistent with their ecological definitions — Forest represents tree canopy while Pasture represents well-maintained grassland with high biomass.

2. Industrial has the lowest positive NDVI (0.18):
   Industrial zones have minimal vegetation, consisting mostly of concrete, metal structures, and paved surfaces. The small positive value comes from occasional patches of grass or trees within industrial boundaries.

3. SeaLake shows strongly negative NDVI (-0.25):
   Open water bodies strongly absorb NIR radiation while reflecting green light, producing negative NDVI values. This perfectly distinguishes water from all land classes and explains why SeaLake achieves perfect classification.

4. PermanentCrop vs AnnualCrop NDVI similarity:
   Both classes show similar NDVI values (0.35 vs 0.38), confirming why they are the most confused pair. Their similar vegetation density makes RGB and NDVI insufficient for distinction — the Red Edge bands are needed instead.

6.3 NDWI Analysis Results

Class	Avg NDWI	Interpretation
SeaLake	+0.44	Strong water presence
River	-0.19	Mixed water and riverbanks
Residential	-0.31	Dry urban surfaces
Industrial	-0.35	Dry industrial surfaces
HerbaceousVegetation	-0.37	Dry vegetation
PermanentCrop	-0.38	Dry crop fields
Pasture	-0.55	Dry grassland
AnnualCrop	-0.55	Dry agricultural land
Forest	-0.45	Low water in canopy
Highway	-0.25	Dry road surfaces

Key Findings from NDWI:

1. SeaLake is the only class with positive NDWI (+0.44):
   This definitively identifies SeaLake as the only class containing significant open water. The strongly positive value confirms that NDWI perfectly isolates water bodies from all land-use classes.

2. River shows slightly negative NDWI (-0.19):
   Unlike SeaLake, River patches contain a mix of water and riverbank pixels at 64×64 resolution. The water pixels push NDWI toward positive while the bank pixels push it negative, resulting in a slightly negative average. This explains why River is occasionally confused with Highway in the RGB model.

3. All land classes show negative NDWI:
   Confirming that no significant standing water exists in agricultural, urban, or vegetation land-use classes.

6.4 Environmental Insights Summary

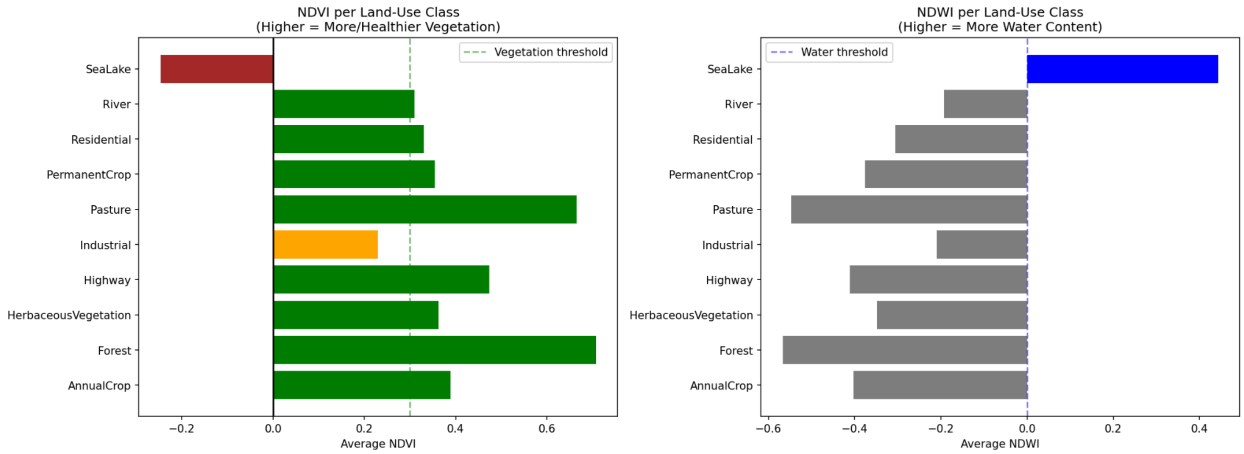
These spectral indices reveal meaningful ecological patterns:

Vegetation Health Gradient:
Forest > Pasture > Highway > AnnualCrop ≈ PermanentCrop ≈ HerbaceousVegetation > Residential > Industrial > SeaLake

This gradient reflects real-world land management intensity: natural ecosystems (Forest) show highest vegetation health while human-modified surfaces (Industrial) show the least.

Water Distribution:
Only SeaLake and River show any meaningful water signal, with SeaLake being definitively water and River being a mixed water-land boundary class.

Urban Heat Island Implication:
Residential and Industrial areas show low NDVI (0.33 and 0.18) and negative NDWI, suggesting reduced vegetation cover and moisture — key factors contributing to urban heat islands. This demonstrates how satellite-based classification can support environmental monitoring and urban planning decisions.



7. Conclusion

7.1 Summary of Achievements

This project successfully built and evaluated two deep learning models for land-use classification from satellite imagery:

Task 1A — RGB Classification:
  Model: EfficientNet-B0 (pretrained on ImageNet)
  Validation Accuracy: 97.80%
  Best Class: Residential, SeaLake (F1: 0.99)
  Weakest Class: PermanentCrop (F1: 0.90)

Task 1B — Multispectral Classification:
  Model: EfficientNet-B0 (modified for 13 channels)
  Validation Accuracy: 98.44%
  Best Class: SeaLake (F1: 1.00 — perfect)
  Weakest Class: Pasture (F1: 0.97)

Task 2 — Explainability:
  Visual: LIME explanations for correct and incorrect predictions
  Spectral: Band ablation study across all 13 Sentinel-2 bands
  Error: Confusion analysis with spectral reasoning

Task 3 — Environmental Insights:
  NDVI: Vegetation health gradient across all 10 classes
  NDWI: Water content analysis identifying SeaLake and River

7.2 Key Conclusions

1. Spectral information improves classification:

The multispectral model outperforms RGB by 0.64% overall, with the biggest gains in visually ambiguous classes like PermanentCrop (+0.07) and HerbaceousVegetation (+0.04). This confirms that wavelengths beyond visible light carry meaningful discriminative information for land-use mapping.

2. NIR is the most critical band:
B8 (Near-Infrared) caused a 29.09% accuracy drop when removed — nearly 3x more than any other band. This aligns with remote sensing principles: NIR strongly differentiates vegetation from water and urban surfaces.

3. Visual ambiguity is the main challenge:
The most confused class pairs (PermanentCrop/AnnualCrop, Pasture/HerbaceousVegetation) are visually identical in RGB but spectrally distinct. This highlights the value of multispectral imaging for fine-grained land classification.

4. Transfer learning is highly effective:
Starting from ImageNet pretrained weights allowed both models to achieve >95% accuracy within just 3 epochs, demonstrating that visual features learned from natural images transfer well to satellite imagery classification.

5. Environmental insights from spectral data:
NDVI and NDWI analysis revealed meaningful ecological patterns — Forest and Pasture have the healthiest vegetation while Industrial areas show the least. SeaLake is uniquely identifiable through its strongly positive NDWI value.

7.3 Limitations

1. Small image size (64×64) limits spatial detail available for classification and reduces Grad-CAM explainability quality, which is why LIME was used instead.

2. Dataset imbalance — Pasture has only 1,400 training images compared to 2,100 for other classes, potentially limiting model performance on this class.

3. Temporal information is not used — satellite images from different seasons would significantly help distinguish AnnualCrop from PermanentCrop based on seasonal changes.

7.4 Future Work

1. Ensemble both RGB and Multispectral models for even higher accuracy by combining their complementary strengths.

2. Use larger input resolution (128×128 or 256×256) for better spatial feature extraction and sharper explainability maps.

3. Incorporate temporal data — using image sequences across seasons would dramatically improve crop type distinction.

4. Explore Vision Transformers (ViT) for multispectral data as they may better capture long-range spatial dependencies in satellite imagery.

7.5 Final Statement

This project demonstrates that combining deep learning with spectral remote sensing principles produces both highly accurate and interpretable land-use classification models. The 98.44% validation accuracy of our multispectral model, combined with comprehensive explainability analysis, provides a strong foundation for real-world geospatial intelligence applications in agriculture monitoring, urban planning, and environmental protection.